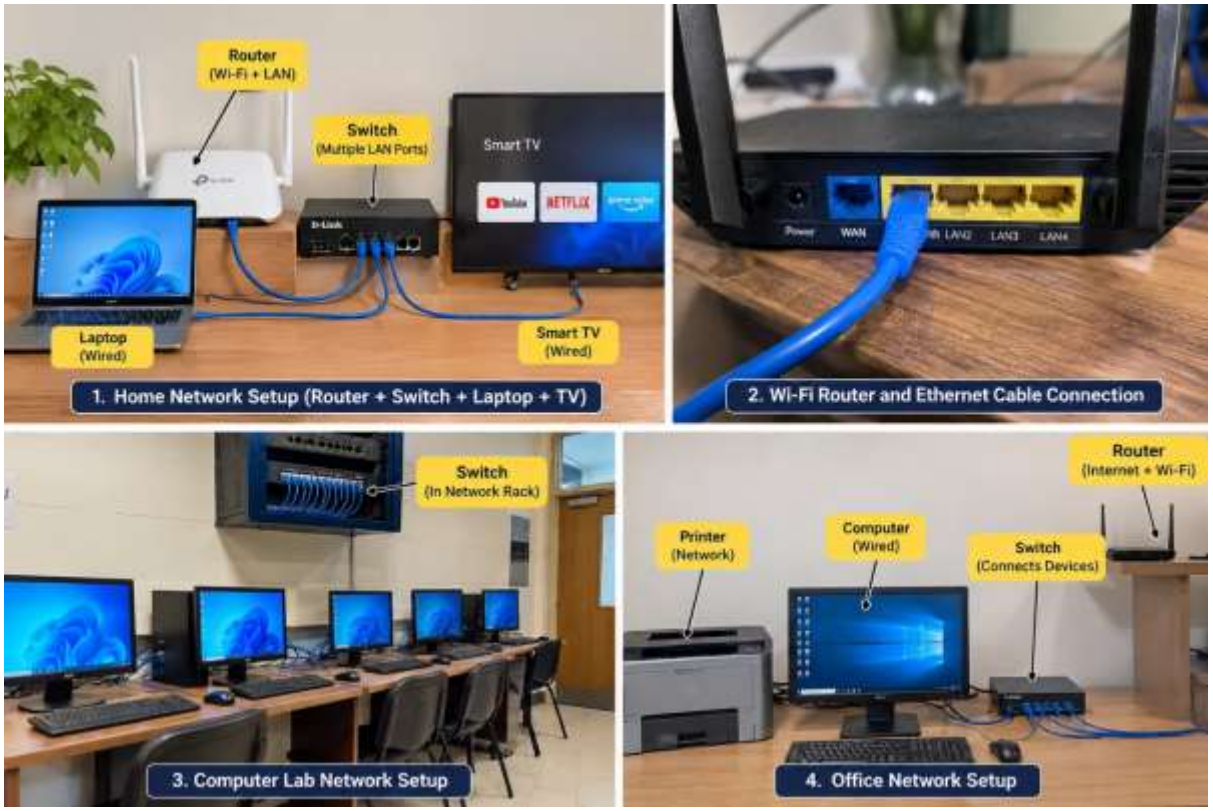
Network Devices

1.Draw and label a simple home network diagram showing a router, switch, and three devices (like a laptop, smartphone, and smart TV), indicating how each device connects to the network.

ANS:



1. Router: Provides internet and Wi-Fi.
2. Switch: Connects multiple devices using Ethernet cables.
3. Laptop: Connects to the switch using an Ethernet cable.
4. Smartphone: Connects to the router using Wi-Fi.
5. Smart TV: Connects to the switch using an Ethernet cable.

2.Create a comparison table listing at least four network **devices (router, switch, hub, access point) and describe one real-world use case for each, referencing apps or services you use daily (e.g., WiFi for streaming on YouTube, LAN gaming, etc.).**

ANS:

Comparison of Network Devices

Network Device	Use	Real-World Example
1. Router	Connects devices to the internet.	Wi-Fi router is used to watch YouTube videos.
2. Switch	Connects multiple computers in one network.	Used in a computer lab for LAN gaming.
3. Hub	Connects multiple computers and sends data to all connected devices.	Used in a small old computer network.
4. Access Point	Provides Wi-Fi to wireless devices.	Used in a college to connect phones and laptops to Wi-Fi.

Simple Explanation

1. Router: Connects us to the internet.
2. Switch: Connects many computers together.
3. Hub: Sends data to all connected computers.
4. Access Point: Provides Wi-Fi connection.

3.Write a short scenario where you explain how data travels from your phone to the internet when you order food on Zomato, mentioning the role of each network device involved (router, switch, access point).

ANS:

1. Access Point: My phone connects to Wi-Fi.
2. Router: The router connects my phone to the internet.
3. Switch: The switch helps send data between network devices.
4. Internet: My order request reaches the Zomato server.
5. Zomato Server: It receives my order and sends confirmation back to my phone.

4.Given this situation: you have a WiFi router, but your gaming PC only has an Ethernet port and is far from the router. Suggest two different network devices you could use to connect your PC to the internet and explain how each would work in this setup.

Hint: Think about devices like switches, powerline adapters, or WiFi extenders.

ANS:

I can use these two devices:
1. Wi-Fi Extender

- It increases the Wi-Fi signal range.

- It receives Wi-Fi from the router and can provide an Ethernet connection to my gaming PC.

2. Powerline Adapter
   - It uses the home's electrical wiring to carry internet signals.
   - I connect one adapter near the router and another near my gaming PC using Ethernet cables.